

MUSTHOFA JOKO ANGGORO

✉ musthofaja.topa@gmail.com | [in linkedin.com/in/musthofa-joko-anggoro](https://www.linkedin.com/in/musthofa-joko-anggoro) | github.com/topahilangharapan | portfolio-topahilangharapan-personal.vercel.app

RESEARCH INTERESTS

My research interests lie in **computer systems** — particularly computer architecture, distributed systems, and operating systems. Through my undergraduate thesis implementing FPGA-based architectures and industry experience optimizing distributed backend systems, I have developed a strong foundation in hardware-software co-design and systems thinking. I am particularly drawn to **edge AI systems** research because it represents a critical challenge at the intersection of resource-constrained computing, energy efficiency, and intelligent workload deployment. I am excited to explore system-level optimizations for AI at the edge, bridging foundational computer architecture with modern distributed systems to enable efficient edge-fog-cloud continuum deployments.

EDUCATION

University of Indonesia (UI) Jakarta, Indonesia
Bachelor of Computer Science, Information Systems *Expected Graduation: Aug 2026*

- **GPA: 3.84/4.00**
- **Thesis Advisor:** Prof. Petrus Mursanto (Full Professor, former Dean of Faculty of Computer Science)
- **Thesis:** Implementation of Morris Mano Computer Architecture with Enhanced Jump Branching Capabilities on Xilinx Spartan-3AN FPGA Platform (XC3S700AN) using VHDL
 - Extended baseline architecture by implementing jump branching in datapath for complex control flow operations
 - Designed and optimized digital logic circuits for hardware-level branching mechanisms
 - Investigated hardware resource utilization and timing constraints for branching logic optimization
 - Synthesized and tested on physical FPGA hardware (not just simulation), validating functional correctness and performance
 - Mainly based on M. Morris Mano's Computer System Architecture
- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

Nanyang Technological University (NTU) & ITS Singapore & Surabaya
INSPIRASI-NTU Summer Program 2025 (LPDP Scholarship Awardee) *July 2025*

- Conducted research on smart port automation, IoT sensor networks, and energy-efficient deployments in distributed maritime systems
- Produced technical report proposing tiered port automation framework with renewable energy utilization and carbon emission reduction

EXPERIENCE

Software Engineer Intern Traveloka (Travel Technology Company)
Systems Engineering & Backend Development *Aug 2025 - Present*

- Resource optimization, distributed system architecture, fault tolerance, cloud infrastructure (AWS, Terraform), containerization (Docker, Kubernetes)

SKILLS

Systems Programming: C, Python, Java, VHDL (hardware description), Go
Hardware & Architecture: FPGA development (Xilinx Spartan-3AN XC3S700AN), VHDL, digital logic design, computer architecture implementation, hardware-software co-design
Machine Learning: Scikit-learn, TensorFlow, supervised/unsupervised learning, model evaluation, imbalanced learning
Distributed Systems: Microservices architecture, containerization (Docker, Kubernetes), cloud infrastructure (AWS, Terraform)
Operating Systems: Linux, Unix — OS internals, process management, memory management, system-level programming

RESEARCH & TECHNICAL PROJECTS

Used Car Sales Prediction System Course Project - AI & Data Science
Systems & ML Integration *Apr 2025 - May 2025*

- Designed end-to-end predictive analytics system integrating multiple ML algorithms (Gradient Boosting, Random Forest, Logistic Regression, K-Means clustering)
- Implemented data preprocessing and feature engineering workflows to handle imbalanced datasets
- Applied performance evaluation methodologies and statistical analysis for model optimization